COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

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Abstract. The Almgren–Pitts combinatorial argument — the cover-refinement plus bookkeeping backbone of Step 2 in min-max constructions of minimal hypersurfaces — is a recurring technical step that each author re-derives in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g), direct minimization over a fixed homology class may produce only degenerate configurations, and the relevant critical points are typically of saddle type. The resolution, following Birkhoff (1917) for closed geodesics and Almgren–Pitts for higher-dimensional varifolds, is a *min-max* construction; regularity of the output is Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set* $\operatorname{Sing}\Sigma$ *has Hausdorff dimension at most $n-7$.*

The core technical result of Pitts's proof — used in every subsequent treatment [CDL03, DLT13, DLR18, DL16, MN14] — is the Almgren–Pitts combinatorial lemma, stated in the Colding–De Lellis formulation as follows.

Proposition 1.2 (Almgren–Pitts combinatorial Lemma, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integral varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Proposition 1.2 reduces to a one-parameter claim on the sweepout [DLT13, §3.2]:

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ ([DLT13, §3.2] Steps 1–2) then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

Scope. This note formalizes Steps 1–2 of the proof of the Claim as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships a single abstract theorem (§2.2) parametric in the sweepout space together with a one-parameter specialization to $[0,1]$ (§2.3); the formalization is machine-verified against `Mathlib`,

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declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

Definition 2.1 (`PairableCover` class, `CombArg.PairableCover`)**.** A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.

Definition 2.2 (`LocalWitness` structure, `CombArg.LocalWitness`)**.** A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

Definition 2.3 (Near-critical set, `CombArg.nearCritical`)**.** For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

Definition 2.4 (`FiniteCoverWithWitnesses` structure, `CombArg.FiniteCoverWithWitnesses`)**.** A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l \in \iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$.

Definition 2.5 (`InitialCover` structure, `CombArg.Refinement.InitialCover`)**.** Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries a finite family of open intervals $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ for $i = 1, \ldots, n$, each paired with a witness center $w_i \in \mathrm{NC}$ and a local witness at w_i with saving $1/(4N)$, subject to (a) spacing: $c_i + r_i < c_{i+2} - r_{i+2}$ for all valid i; (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: $\mathrm{NC} \subseteq \bigcup_i I_i$.[1]

Definition 2.6 (`PartialRefinement` structure, `CombArg.Refinement.PartialRefinement`)**.** Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k.

2.2. Public API: abstract core.

Theorem 2.7 (Abstract core, `CombArg.exists_sup_reduction_of_cover`)**.**
Input. *A compact, nonempty topological space K; a continuous $f : K \to \mathbb{R}$; $m_0 = \sup f$; scalars $0 < \varepsilon \leq \delta$; a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ (Definition 2.4).*
Output. *There exists $f' : K \to \mathbb{R}$ with $\sup f' \leq m_0 - \varepsilon$.*

2.3. Public API: one-parameter application.

Theorem 2.8 (One-parameter application, `CombArg.exists_sup_reduction`)**.**
Input. *A pseudo-metric space X with a `PairableCover X` instance (Definition 2.1); a continuous $f : \mathbf{I} \to \mathbb{R}$; $m_0 \in \mathbb{R}$ with $m_0 > 0$ and $m_0 = \sup f$; $N \in \mathbb{N}$ with $N > 0$; for every $t \in \mathrm{NC}$ a `LocalWitness` at t with saving $1/(4N)$ (Definition 2.2).*
Output. *There exists $f' : \mathbf{I} \to \mathbb{R}$ with $\sup f' \leq m_0 - 1/(4N)$.*
Specializes Theorem 2.7 to $K = \mathbf{I}$, $\delta = 1/N$, $\varepsilon = 1/(4N)$.

[1] Conditions (a) and (b) transcribe [DLT13, §3.2]'s spacing and subset conditions; the pair (c_i, w_i) separates interval centers from witness centers, required because grid points need not themselves be near-critical.

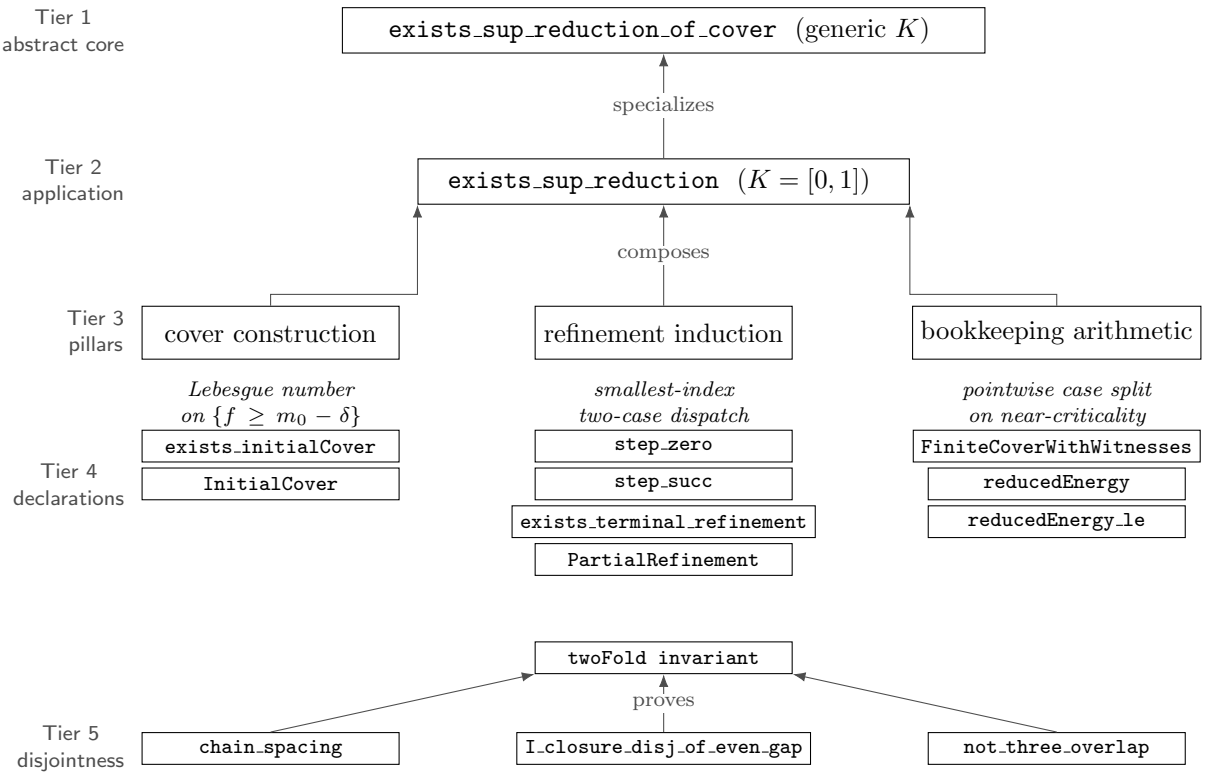

FIGURE 1. `CombArg` v0.1 architecture. The abstract core (Tier 1) is parametric in the sweepout space; the one-parameter application (Tier 2) instantiates $K = [0, 1]$ and composes three construction pillars (Tier 3). Each pillar's key declarations appear in Tier 4. The disjointness sub-module (Tier 5) supplies the two-fold overlap invariant consumed by bookkeeping.

2.4. **Internal lemmas.** The declarations below are consumed by Theorem 2.8; proofs are sketched in §3.

Proposition 2.9 (Compactness of NC, `CombArg.Refinement.isCompact_nearCritical`, `CombArg.Refinement.n`
If K is compact and f continuous, NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty.*

- `CombArg.Refinement.exists_initialCover` — existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- `CombArg.Refinement.exists_terminal_refinement` — existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- `CombArg.Refinement.InitialCover.not_three_overlap` — at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- `CombArg.Refinement.saving_bound_closure` — the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.
- `CombArg.Refinement.exists_refinement` — packages the 1D cover construction output as a `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$, the input to Theorem 2.7.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.8 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and deriving TwoFold. Theorem 2.7 closes.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover, `CombArg.Refinement.exists_initialCover`)**.** *Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.5) exists.*

Construction. Compactness of NC (Proposition 2.9) plus the open cover $\{U_t\}_{t\in\mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \ldots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case, `CombArg.Refinement.step_zero`)**.** step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

Definition 3.3 (Inductive step, `CombArg.Refinement.step_succ_at`)**.** Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \ldots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[2]

Proposition 3.4 (Termination, `CombArg.Refinement.exists_terminal_refinement`)**.** *Starting from* step_zero(ic) *and iterating* step_succ_at*, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

Argument. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in \mathrm{rem}(\mathrm{pr})$ (chosen as the smallest such) produces pr′ with $i^* \notin \mathrm{rem}(\mathrm{pr}')$ and $\mathrm{rem}(\mathrm{pr}') \subseteq \mathrm{rem}(\mathrm{pr}) \setminus \{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell + 1) = i^*$ is fresh because $i^* \in \mathrm{rem}$ while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. Stage C: assembly.

The final stage extends the saving bound from open J_k to $\overline{J_k}$ and derives TwoFold from spacing (a).

Proposition 3.5 (TwoFold via parity, `CombArg.Refinement.terminal_twoFold`)**.** *For a partial refinement* pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \ldots, \overline{J_L}$.*

[2] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

Argument. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i - j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (`CombArg.Refinement.InitialCover.chain_spacing`, `CombArg.Refinement.InitialCover.closure_disjoint`

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \ldots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \ldots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (`CombArg.Refinement.InitialCover.not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound, `CombArg.Refinement.saving_bound_closure`)**.** *Let* pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Argument. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Lemma 3.7 (Assembly, `CombArg.Refinement.exists_refinement`)**.** *A partial refinement* pr *of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i yields a refinement of $(\mathbf{I}, X, f, m_0, N)$ in the sense of Definition 2.4, with $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and $E_k :=$ the replacement energy of the $\sigma(k)$-th witness.*

Argument. Invariant (I) is Proposition 3.6. Invariant (II) is Proposition 3.5. Invariant (III) holds by construction ($s_k = 1/(4N)$). Coverage of NC chains $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Theorem 2.8 now follows: chain Propositions 3.1 and 3.4, invoke Lemma 3.7, and apply Theorem 2.7.

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